

SOC-5811 Week 11: Panel Data

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REVIEW: INTERACTIONS

$$\text{PolViews}_i = \beta_0 + \beta_1 \text{Religion}_i + \beta_2 \text{Age}_i$$



REVIEW: INTERACTIONS

$$\text{PolViews}_i = \beta_0 + \beta_1 \text{Religion}_i + \beta_2 \text{Age}_i + \beta_3 (\text{Religion}_i \times \text{Age}_i)$$



REVIEW: INTERACTIONS

$$\text{PolViews}_i = \beta_0 + \boxed{\beta_1} \text{Religion}_i + \beta_2 \text{Age}_i + \beta_3 (\text{Religion}_i \times \text{Age}_i)$$

REVIEW

- What do “fixed effects” do in linear regression models fit to panel data?

$$y_{i,t} = \beta_0 + \beta_1 X_{i,t} + \alpha_i$$

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- ▶ Basically the same thing as controlling for a categorical variable, but the “category” is units.
 - ▶ Intercept shift by unit.
 - ▶ Isolates our target comparison to be based on average *within-unit* variation (i.e., variation over time).
 - ▶ From a causal perspective, eliminates all **time-invariant** confounders.

REVIEW

- ▶ (1) Unit-level fixed effects cannot address **time-variant** unobserved confounding.
- ▶ (2) Unit-level fixed effects can lead to substantial bias if there is strong reverse causality.
- ▶ (3) Including fixed effects isn't just a "better" way to estimate the same comparison; it *changes* the comparison.



REVIEW

- ▶ When do I want to estimate coefficients using **between-unit** differences vs. **within-unit** differences?



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- ▶ If you care about estimating **causal effects** (or something you want to generally interpret that way), you probably want to use within-unit differences.



REVIEW

- ▶ When do I want to estimate coefficients using **between-unit** differences vs. **within-unit** differences?
- ▶ If you care about estimating **causal effects** (or something you want to generally interpret that way), you probably want to use within-unit differences.
- ▶ If you care about estimating **descriptive disparities**, you probably want to use between-unit differences.



BETWEEN VS. WITHIN

$i = \text{hospital}$

$t = \text{year}$

$$\text{Physicians}_{i,t} = \beta_0 + \beta_1 \text{SafetyNet}_{i,t} + \beta_2 Z$$

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BETWEEN VS. WITHIN

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BETWEEN VS. WITHIN

- **Research question:** Do safety net hospitals have different average physicians per patient, controlling for Z ?

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BETWEEN VS. WITHIN

- ▶ **Research question:** Do safety net hospitals have different average physicians per patient, controlling for Z ?
- ▶ β_1 : Across all hospitals, safety net hospitals have β_1 more/fewer physicians per capita.

$$\text{Physicians}_{i,t} = \beta_0 + \boxed{\beta_1} \text{SafetyNet}_{i,t} + \beta_2 Z$$

BETWEEN VS. WITHIN

- ▶ **Research question:** Do safety net hospitals have different average physicians per patient, controlling for Z ?
- ▶ β_1 : Across all hospitals, safety net hospitals have β_1 more/fewer physicians per capita.
- ▶ Descriptive comparison with implications for stratification/inequality.

$$\text{Physicians}_{i,t} = \beta_0 + \boxed{\beta_1} \text{SafetyNet}_{i,t} + \beta_2 Z$$

BETWEEN VS. WITHIN

- **Research question:** As the patient population changes within hospitals, does average physicians per patient change?

$$\text{Physicians}_{i,t} = \beta_0 + \boxed{\beta_1} \text{SafetyNet}_{i,t} + \beta_2 Z + \alpha_i$$

BETWEEN VS. WITHIN

- ▶ **Research question:** As the patient population changes within hospitals, does average physicians per patient change?
- ▶ Causal comparison with implications for the effect of patient population on work force within hospitals. When hospitals become safety net hospitals, do they see reductions in work force?

$$\text{Physicians}_{i,t} = \beta_0 + \boxed{\beta_1} \text{SafetyNet}_{i,t} + \beta_2 Z + \alpha_i$$

BETWEEN VS. WITHIN

- ▶ **Research question:** Do safety net hospitals have different average physicians per patient, controlling for Z ?
- ▶ **Research question:** As the patient population changes within hospitals, does average physicians per patient change?
- ▶ **Both answer interesting but different research questions.**



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- ▶ Fixed effects estimators (modified OLS): **`feols()`**.

ESTIMATION

```
summary(lm(rate ~ beertax + state, data=Fatalities))
```

```
##
## Call:
## lm(formula = rate ~ beertax + state, data = Fatalities)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.8696 -0.8284 -0.0127  0.7955  8.9780
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   34.7763     3.1336  11.098 < 2e-16 ***
## beertax       -6.5587     1.8785  -3.491 0.000556 ***
## stateaz       -5.6773     2.6667  -2.129 0.034107 *
## statear       -6.5495     2.1902  -2.990 0.003028 **
## stateca      -15.0947     3.0435  -4.960 1.21e-06 ***
## stateco      -14.8428     2.8735  -5.165 4.50e-07 ***
## statect      -18.6226     2.8053  -6.638 1.58e-10 ***
## statede      -13.0760     2.9395  -4.448 1.24e-05 ***
## statefl      -2.6813     1.3933  -1.924 0.055284 .
## statega       5.2460     1.8395   2.852 0.004661 **
## stateid      -6.6902     2.5797  -2.593 0.009989 **
## stateil      -19.6162     2.9150  -6.730 9.23e-11 ***
## statein      -14.6154     2.7254  -5.363 1.69e-07 ***
## stateia      -15.4393     2.5344  -6.092 3.58e-09 ***
## stateks      -12.2322     2.4544  -4.984 1.08e-06 ***
## stateky      -12.1752     2.8717  -4.240 3.02e-05 ***
## statela      -8.4712     1.8867  -4.490 1.03e-05 ***
## statemi      -11.0785     1.9112  -5.797 1.78e-08 ***
## statemn      -17.0644     2.8322  -6.025 5.17e-09 ***
## statems       2.7610     2.7610   1.000 0.324e-13 ***
```

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ESTIMATION

```
summary(lm(rate_demeaned ~ tax_demeaned, data=Fatalities))
```

```
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##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.8696 -0.8284 -0.0127  0.7955  8.9780
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.348e-16  9.601e-02   0.000 1.000000
## tax_demeaned -6.559e+00  1.741e+00  -3.767 0.000196 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.76 on 334 degrees of freedom
## Multiple R-squared:  0.04074,    Adjusted R-squared:  0.03787
## F-statistic: 14.19 on 1 and 334 DF,  p-value: 0.0001956
```

```
summary(feols(rate ~ beertax | state, data=Fatalities))
```

```
## OLS estimation, Dep. Var.: rate
## Observations: 336
## Fixed-effects: state: 48
## Standard-errors: Clustered (state)
##           Estimate Std. Error  t value Pr(>|t|)
## beertax -6.55874      2.91856 -2.24725 0.029358 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.'
## RMSE: 1.7547      Adj. R2: 0.889129
##           Within R2: 0.040745
```

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- ▶ Fixed effects estimators (modified OLS): **`feols()`**.

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- ▶ OLS estimator: Add unit as a categorical variable in `lm()`.
 - ▶ Very slow/clunky if you have thousands of units.
- ▶ OLS estimator: Manually demean and then `lm()` with demeaned data.
 - ▶ Very complex to do correctly by hand if you have many variables in your model.
- ▶ Fixed effects estimators (modified OLS): **feols()**.
 - ▶ Very fast and efficient; the tradeoff is that you don't estimate standard errors on your fixed effects themselves, but typically this does not matter for your target comparison.

UNCERTAINTY

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- ▶ Remember that in the conventional regression model, we assume errors are **independent**. In sampling terms, we assume every sample from the population model is independent of any other sample.
- ▶ Think about flipping a coin 100 times. When we estimate the standard error of the random variable X = number of heads in 100 flips, we assume each flip is unrelated to (or independent from) the other flips.

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- ▶ Think about flipping a coin 100 times. When we estimate the standard error of the random variable X = number of heads in 100 flips, we assume each flip is unrelated to (or independent from) the other flips.
- ▶ Solution: **clustered standard errors**.

UNCERTAINTY

β_1 from our basic OLS model with unit as a categorical variable:

##	Estimate	Std. Error	t value	Pr(> t)
##	-6.5587372215	1.8784999358	-3.4914758828	0.00055596

β_1 from our fixed effects model:

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## OLS estimation, Dep. Var.: rate
## Observations: 336
## Fixed-effects: state: 48
## Standard-errors: Clustered (state)
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REVIEW

Pick up here on Wednesday.



INTERACTIONS REVIEW

Categorical $\times$ Categorical

$$\text{MPG}_i = \beta_0 + \beta_1 E_i + \beta_2 T_i + \beta_3 (E_i \times T_i)$$

Categorical $\times$ Categorical

$$\text{MPG}_i = \beta_0 + \beta_1 E_i^{\text{v-shaped}} + \beta_2 T_i^{\text{manual}} + \beta_3 (E_i^{\text{v-shaped}} \times T_i^{\text{manual}})$$

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```
##
## Call:
## lm(formula = mpg ~ engine_type * transmission, data = mtcars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.971 -1.973  0.300   2.036   6.250
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   20.743     1.312   15.808 1.75e-15 ***
## engine_typev-shaped            -5.693     1.651   -3.448 0.001805 **
## transmissionmanual              7.629     1.856    4.111 0.000312 ***
## engine_typev-shaped:transmissionmanual -2.929     2.541   -1.153 0.258855
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.472 on 28 degrees of freedom
## Multiple R-squared:  0.7003, Adjusted R-squared:  0.6682
## F-statistic: 21.81 on 3 and 28 DF,  p-value: 1.735e-07
```

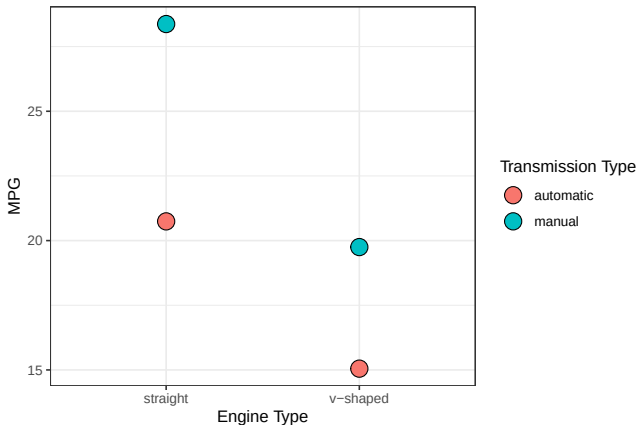
Categorical $\times$ Categorical

$$\text{MPG}_i = (20.7) + (-5.7)E_i^{\text{v-shaped}} + (7.6)T_i^{\text{manual}} + (-2.9)(E_i^{\text{v-shaped}} \times T_i^{\text{manual}})$$

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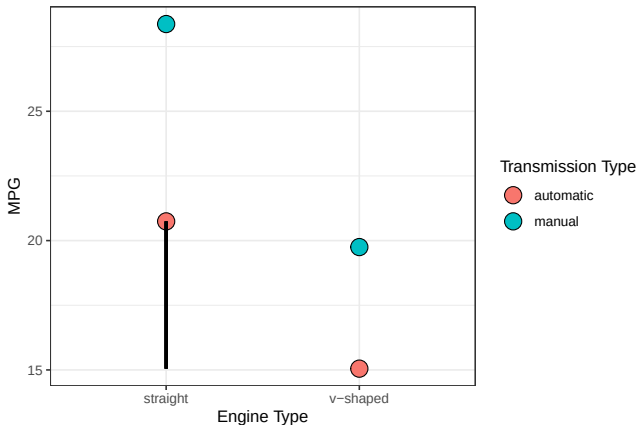
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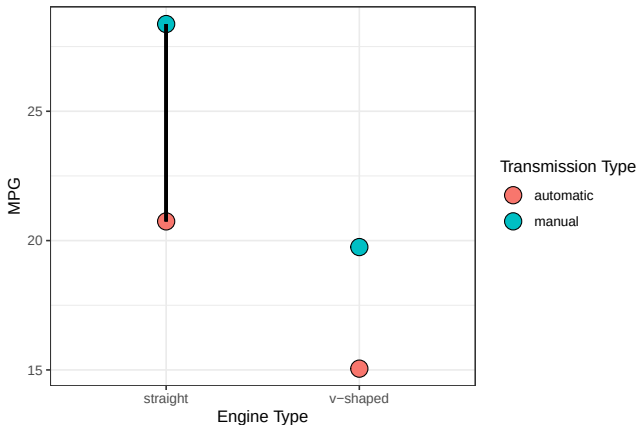
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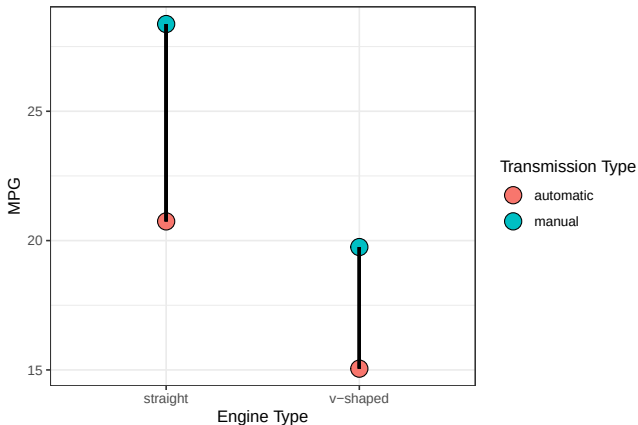
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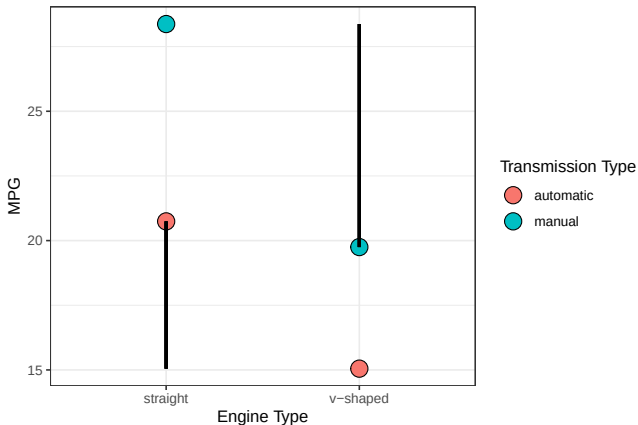
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- ▶ β_1 : Across all cars at the reference level for Transmission (Transmission=automatic), v-shaped Engines are associated with -5.7 lower MPG.
- ▶ β_2 : Across all cars at the reference level for Engine (Engine=straight), manual Transmissions are associated with 7.6 higher MPG.
- ▶ β_3 : The association between v-shaped Engines and MPG for Transmission=automatic (-5.7) is -2.9 lower for Transmission=manual. The association between manual Transmissions and MPG for Engine=straight (7.6) is -2.9 lower for Engine=v-shaped.

INTERACTIONS REVIEW

Continuous $\times$ Categorical

$$\text{MPG}_i = \beta_0 + \beta_1 H_i + \beta_2 C_i + \beta_3 (H_i \times C_i)$$

Continuous $\times$ Categorical

$$\text{MPG}_i = \beta_0 + \beta_1 \text{H}_i + \beta_2 \text{C}_i^6 + \beta_3 \text{C}_i^8 + \beta_4 (\text{H}_i \times \text{C}_i^6) + \beta_5 (\text{H}_i \times \text{C}_i^8)$$

Continuous $\times$ Categorical

$$\text{MPG}_i = \beta_0 + \beta_1 H_i + \beta_2 C_i^6 + \beta_3 C_i^8 + \beta_4 (H_i \times C_i^6) + \beta_5 (H_i \times C_i^8)$$

```
##
## Call:
## lm(formula = mpg ~ hp * as.factor(cyl), data = mtcars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.7600 -1.8152 -0.1971  1.5012  7.1606
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    35.98303     3.88908   9.252 1.04e-09 ***
## hp             -0.11278     0.04575  -2.465  0.02061 *
## as.factor(cyl)6 -15.30917     7.43456  -2.059  0.04962 *
## as.factor(cyl)8 -17.90295     5.25961  -3.404  0.00216 **
## hp:as.factor(cyl)6  0.10516     0.06848   1.536  0.13672
## hp:as.factor(cyl)8  0.09853     0.04862   2.026  0.05310 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.028 on 26 degrees of freedom
## Multiple R-squared:  0.7882, Adjusted R-squared:  0.7475
## F-statistic: 19.35 on 5 and 26 DF,  p-value: 5.019e-08
```

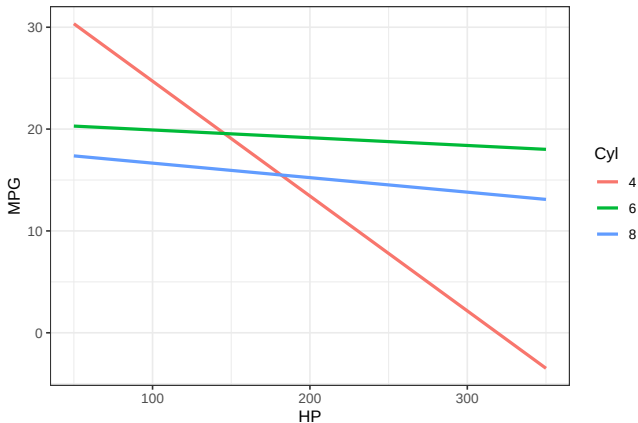
Continuous $\times$ Categorical

$$\text{MPG}_i = (36) + (-0.1)H_i + (-15)C_i^6 + (-18)C_i^8 + (0.1)(H_i \times C_i^6) + (0.1)(H_i \times C_i^8)$$

```
##  
## Call:  
## lm(formula = mpg ~ hp * as.factor(cyl), data = mtcars)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -4.7600 -1.8152 -0.1971  1.5012  7.1606   
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Continuous $\times$ Categorical

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- ▶ β_1 : Across all cars at the reference level for Cylinder (Cylinder=4), a 1 unit increase in Horsepower is associated with -0.1 lower MPG.
- ▶ β_2 : Across all cars at the reference level for Horsepower (Horsepower=0), Cylinder=6 is associated with -15 lower MPG.
- ▶ β_3 : Across all cars at the reference level for Horsepower (Horsepower=0), Cylinder=8 is associated with -18 lower MPG.
- ▶ β_4 : The association between Horsepower and MPG for Cylinder=4 (-0.1) is 0.1 higher for Cylinder=6. The association between Cylinder=6 and MPG for Horsepower=0 (-15) is 0.1 higher for every 1 unit increase in Horsepower.
- ▶ β_5 : The association between Horsepower and MPG for Cylinder=4 (-0.1) is 0.1 higher for Cylinder=8. The association between Cylinder=8 and MPG for Horsepower=0 (-18) is 0.1 higher for every 1 unit increase in Horsepower.

INTERACTIONS REVIEW

Continuous $\times$ Continuous

$$\text{MPG}_i = \beta_0 + \beta_1 H_i + \beta_2 W_i + \beta_3 (H_i \times W_i)$$

Continuous $\times$ Continuous

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```
##
## Call:
## lm(formula = mpg ~ hp * wt, data = mtcars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.0632 -1.6491 -0.7362  1.4211  4.5513
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  49.80842    3.60516   13.816 5.01e-14 ***
## hp          -0.12010    0.02470   -4.863 4.04e-05 ***
## wt          -8.21662    1.26971   -6.471 5.20e-07 ***
## hp:wt         0.02785    0.00742    3.753 0.000811 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.153 on 28 degrees of freedom
## Multiple R-squared:  0.8848, Adjusted R-squared:  0.8724
## F-statistic: 71.66 on 3 and 28 DF,  p-value: 2.981e-13
```

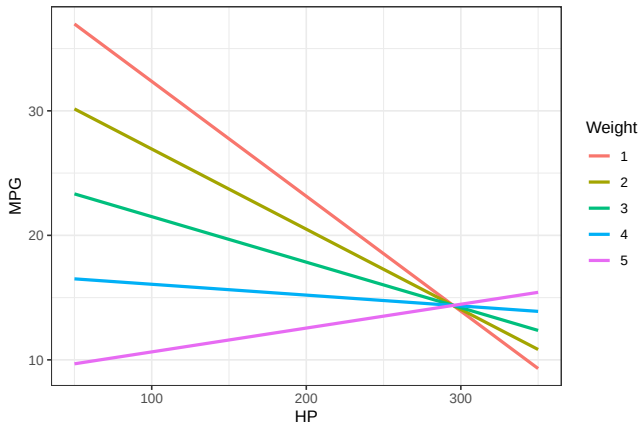
Continuous $\times$ Continuous

$$\text{MPG}_i = (49.8) + (-0.12)H_i + (-8.2)W_i + (0.02)(H_i \times W_i)$$

```
##
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- ▶ β_1 : Across all cars at the reference level for Weight (Weight=0), a 1 unit increase in Horsepower is associated with -0.12 lower MPG.
- ▶ β_2 : Across all cars at the reference level for Horsepower (Horsepower=0), a 1 unit increase in Weight is associated with -8.2 lower MPG.
- ▶ β_3 : The association between Horsepower and MPG for Weight=0 (-0.12) is 0.02 higher for every 1 unit increase in Weight. The association between Weight and MPG for Horsepower=0 (-8.2) is 0.02 higher for every 1 unit increase in Horsepower.

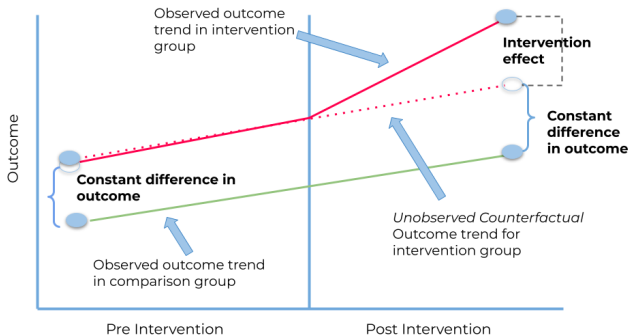
TWO-WAY FIXED EFFECTS

- Two-way fixed effects and brief introduction to difference-in-differences.

```
summary(feols(rate~beertax | state+year, data=Fatalities))
```

```
## OLS estimation, Dep. Var.: rate
## Observations: 336
## Fixed-effects: state: 48, year: 7
## Standard-errors: Clustered (state)
##      Estimate Std. Error  t value Pr(>|t|)
## beertax  -6.3998    3.57078 -1.79227 0.079528 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 1.71819      Adj. R2: 0.891425
##                Within R2: 0.036065
```

TWO-WAY FIXED EFFECTS



REVIEW: FIXED EFFECTS

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 - ▶ Intercept shift by unit.
 - ▶ Isolates our target comparison to be based on average *within-unit* variation (i.e., variation over time).
 - ▶ From a causal perspective, eliminates all **time-invariant** confounders.

REVIEW: FIXED EFFECTS

Notation:

$$y_{i,t} = \beta_0 + \beta_1 X_{i,t} + \beta_2 Z + \alpha_i$$

Estimation:

- Fixed effects estimators (modified OLS): **feols()**.

```
feols(rate ~ beertax | state, data=Fatalities)
```

REVIEW: FIXED EFFECTS

Caveats:

- ▶ (1) Unit-level fixed effects cannot address **time-variant** unobserved confounding.
- ▶ (2) Unit-level fixed effects can lead to substantial bias if there is strong reverse causality.
- ▶ (3) Including fixed effects isn't just a "better" way to estimate the same comparison; it *changes* the comparison.